

Response Letter with Annotations

Zhang, Y., Lukito, J., Su, M.-H., Suk, J., Xia, Y., Kim, S. J., Doroshenko, L., & Wells, C. (2021). Assembling the Networks and Audiences of Disinformation: How Successful Russian IRA Twitter Accounts Built Their Followings, 2015–2017. Journal of Communication, 71(2), 305–331. <https://doi.org/10.1093/joc/jgaa042>

Annotations

1. Purposes

a. Motivation (one-sentence summary)

This study aimed to understand how Russian Internet Research Agency (IRA) Twitter accounts constructed followers in a politically polarized society and media ecosystem around the 2016 U.S. presidential election because the IRA twitter followings likely contributed to the huge success of IRA disinformation campaigns.

b. Specific Rationale (several bullet points)

- The study conceptualizes social media following as both an **ego network** and as a (direct) audience to explain its communicative power.

***Ego network:** the set of individuals with which an account has direct ties.

- Twitter followings facilitated the direct dissemination of strategic messages by serving as their audiences.
- Followings (including the ego network) served as a set of individuals amplifying messages from the IRA through retweeting and other means. It leads to “multiplier effect” that strengthens the influence of central actors (IRA accounts in this case), because a large follower base enables access to more networks of social ties, resources, and influence.
- An ego network allows social media accounts to construct more followers by “transitivity of attention” and social recommendation, in which nonfollowers are exposed to messages retweeted by their contacts (Golder & Yardi, 2010).

- The high number of followings of IRA accounts served as visible engagement signals to observers, including journalists, showing authority and authenticity.
- **Network effects in social media: Homophilous networks (bottom-up network effect):**

Individuals are more likely to disseminate information shared by like-minded peers (Aral et al., 2009).

As IRA tweets are retweeted through a homogenous network, endorsement from within the network may facilitate growth in its following.

- **Influential actors (Top-down network effect):**

The two-step flow theory suggests that information is diffused by opinion leaders” or “influentials” first (Katz et al., 2017).

Influential actors’ retweeting of IRA accounts may influence IRA accounts’ followings.

- **News media amplification**

News media leverages social media to attract audiences in the competitive and fragmented news ecology.

This introduction of social media from news media amplifies social media.

Partisan news media:

Partisan news media amplify ideologically consistent social media content, leading to more exposure of social media and ultimately increase in followers of those accounts.

Mainstream media:

Mainstream media could also recirculate disinformation by embedding IRA accounts’ tweets in their news stories, for example.

Likewise, it would introduce IRA social media accounts to the audience, which would ultimately increase followers of those accounts.

c. Research Questions and Hypotheses

H1: The more retweets IRA accounts gain, the more followers they accumulate.

*IV: retweets that IRA accounts gain, DV: followers that IRA accounts gain

H2: The greater is the size of follower networks of those retweeting IRA accounts, the more follower's IRA accounts gain.

*IV: the size of follower networks of those retweeting IRA accounts, DV: followers that IRA accounts gain

H3: Influential actors' retweeting of IRA accounts will increase IRA accounts' followings.

*IV: influential actors' retweeting of IRA accounts, DV: followers that IRA accounts gain

H4: Like-minded partisan media's uptake of IRA tweets leads to an increase in IRA accounts' followers.

*IV: like-minded partisan media's uptake of IRA tweets, DV: followers that IRA accounts gain

H5: Mainstream media's uptake of IRA tweets leads to an increase in IRA accounts' followers.

*IV: mainstream media's uptake of IRA tweets, DV: followers that IRA accounts gain

RQ1: Who retweeted the IRA accounts?

2. Method

a. Research Design

- **Quantitative case study using timeseries analysis modeling.**

The study stated that it took a quantitative case study approach and analyzed how the four most successful English-speaking IRA Twitter accounts (quantified in terms of retweets; Cha et al., 2010) constructed their followings. Furthermore, they conducted timeseries analysis on variables to see the relationship among variables (more explanations in Analysis part).

b. Sample

- **Non representative sample; convenient sample.**

IRA-related tweets:

The study selected four most successful IRA twitter accounts, two of which masqueraded as conservatives and other two masqueraded as liberals. (They collected all tweets that turned out to be posted by or retweeted the 2,752 Russian IRA accounts between 2015 and 2017 from a Twitter archive that randomly collects 1% of Global Twitter stream and selected those four accounts.)

News media:

The study retained media outlets that referenced or quoted the IRA accounts and filtered that had a stable stream of publications in MediaCloud, and classified them into mainstream, hyperliberal, and hyperconservative based on existing classifications in those collections.

A sample of Twitter users who retweeted the IRA accounts:

They used data collection of tweets from 375,725 accounts who followed Donald Trump on Election Day 2016, which was 3% of his following at that time. Each account was categorized using spectral clustering based on similarity in who those accounts followed. The study focused on the six politically active categories of accounts: “Trump supporters,” “Far right conservatives,” “Alt- Right,” “Mainstream conservatives,” “Liberals,” and “Mainstream politics.” (The study states that these accounts following Trump were not relevant to the present analysis!)

c. Measures

Follower count:

They obtained follower counts for IRA accounts from tweet metadata, along with records of other users' retweets of IRA posts, using the 1% Twitter data stream. When multiple follower count values appeared on the same day, those values were averaged the follower counts by day.

Number of retweets:

They computed the daily number of retweets by counting how many times each IRA account was retweeted by other Twitter accounts that were observed in the 1% sample.

Size of retweet network:

The study defined an account's daily retweet network as the users who retweeted a given IRA account per day. The "size of retweet network" was calculated as the sum of those users' follower counts, representing the account's daily potential exposure from retweets despite possible audience overlap.

Retweeting by influential actors:

The study identified all verified Twitter users in the 1% sample who retweeted the IRA accounts and treated them as influential actors.

News media coverage:

The study obtained the number of articles referencing an IRA account each day by media type which are hyperliberal, mainstream, and hyperconservative.

d. Analysis

- Time series modeling using the vector autoregression (VAR) model including Wald two-way Granger causality tests and Impulse response functions (IRFs).

- Vector autoregression (VAR) was used to determine the time-ordered relationship between variables. (*Variables: retweet count, size of retweet network, number of retweeting by influential actors, number of hyperconservative media stories, number of hyperliberal media stories, number of mainstream media stories, and follower count.)
- To account for all potential relationships, they treated every variable as endogenous and examined the relationship bidirectionally.

**Endogenous variable: Simply put, variable that can be determined and explained in certain model*

**VAR captures the interdependencies between endogenous variables by fitting a multivariate time series regression of each endogenous variable on lags of itself and lags of all other endogenous*

**Wald two-way Granger causality test is a bivariate test of causality.*

**Impulse response functions (IRFs) from the VAR models provide further information regarding the longer-term effects of one variable (X) on another (Y), by testing the impact or “shock” of X on Y (Swanson & Granger, 1997).*

3. Results and Conclusions

a. Answers to RQs and Hypotheses

H1: The more retweets IRA accounts gain, the more followers they accumulate.

H1 was supported: Wald two-way Granger causality tests and IRFs show that across all four IRA accounts, the number of retweets lead to follower increase.

H2: The greater is the size of follower networks of those retweeting IRA accounts, the more follower’s IRA accounts gain.

H2 was partially supported: IRFs partially confirm the effect of the retweet network size (H2). For @TEN_GOP and @Crystal1Johnson, H2 was supported. However, not statistically significant for @Pamela_Moore13 and @gloed_up.

H3: Influential actors’ retweeting of IRA accounts would increase IRA accounts’ Twitter followings.

H3 was partially supported: Granger causality tests indicate that retweeting of IRA messages by influential actors was associated with follower growth for @TEN_GOP and

@Crystal1Johnson, but IRF analysis shows statistically significance only for @TEN_GOP.

H4: Like-minded partisan media's uptake of IRA tweets leads to an increase in IRA accounts' followers.

H4 was not supported: Granger causality tests suggest that hyperconservative media quoting increased followers for the two conservative IRA accounts, but IRF analyses show no statistically significant direct effect after controlling for other factors, implying a mediated influence.

H5: Mainstream media's uptake of IRA tweets leads to an increase in IRA accounts' followers.

H5 was partially supported: Mainstream media's uptake of two conservative tweet accounts Granger caused their followings to increase, with IRFs confirming this result. Furthermore, Granger causality tests suggested that hyperconservative media's effect on their follower growth was mediated by mainstream media coverage. However, any media uptake including mainstream media did not have any effect on these two liberal accounts' follower growth.

RQ1: Who retweeted IRA accounts?

The partisan retweeting of the four IRA accounts' tweets was obvious. Far-right conservatives and the alt-right together made up over 70% of the sample who retweeted @TEN_GOP and contributed to over 80% of the retweets while liberals accounted for about 60% of the sample who retweeted @Crystal1Johnson and @gloed_up and contributed to around 70% of the retweets.

b. Include Key Figures/Tables

Table 1 Two-Way Granger Causality Wald tests

Relationship	Conservative Accounts				Liberal Accounts			
	@TEN_GOP		@Pamela_Moore13		@Crystal1Johnson		@gloed_up	
	F	p-value	F	p-value	F	p-value	F	p-value
Number of retweets → followers	58.726	0.000***	104.280	0.000***	66.932	0.000***	6.979	0.008**
Size of retweet network → followers	45.783	0.000***	58.441	0.000***	52.654	0.000***	5.604	0.018*
Retweeting by influential actors → followers	38.950	0.000***	0.453	0.502	13.838	0.000***	0.455	0.500
Hyperconservative media uptake → followers	20.832	0.000***	6.439	0.012*	0.242	0.623	0.610	0.435
Hyperliberal media uptake → followers	9.436	0.002**	0.000	0.988	0.863	0.353	0.141	0.708
Mainstream media uptake → followers	10.100	0.002**	4.981	0.026*	0.572	0.450	0.045	0.832
Followers → number of retweets	0.686	0.408	0.261	0.610	0.000	0.983	0.124	0.725
Followers → size of retweet network	1.107	0.293	0.231	0.631	0.401	0.527	0.151	0.698
Followers → retweeting by influential actors	0.135	0.713	0.018	0.894	0.260	0.611	0.000	0.993
Followers → hyperconservative media uptake	0.476	0.490	0.268	0.605	0.033	0.856	0.001	0.971
Followers → hyperliberal media uptake	0.084	0.773	0.009	0.924	0.087	0.768	0.012	0.914
Followers → mainstream media uptake	0.298	0.586	0.066	0.798	0.001	0.982	0.000	0.988

* $p < .05$, ** $p < .01$, *** $p < .001$

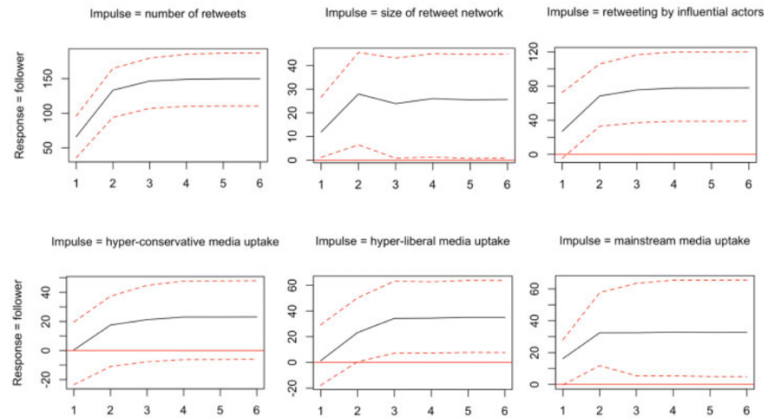


Figure 2 IRFs for @TEN_GOP (conservative), 95% bootstrap confidence interval (dotted ribbon), 100 runs. Effects are cumulative.

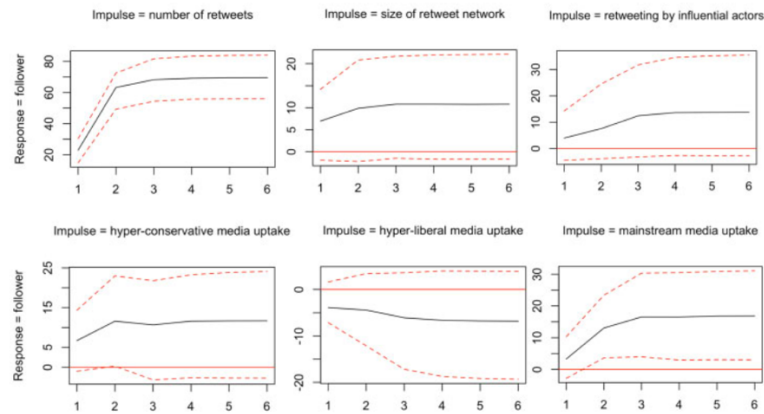


Figure 3 IRFs for @Pamela_Moore13 (conservative), 95% bootstrap confidence interval (dotted ribbon), 100 runs. Effects are cumulative.

c. Authors' Conclusions

- Partisan individuals on social media increased IRA accounts' influence by amplifying the disinformation through retweeting.
- News media, driven by the attention-seeking impulse, increased IRA accounts' influence by embedding IRA tweets directly in news.
- IRA took advantage of these weaknesses and develop unsuspecting followings.
- The mechanisms for accumulation of followers were asymmetrical for the conservative and liberal IRA accounts: Liberal IRA accounts relied on partisan social media networks, while conservative IRA accounts activated the entire conservative media ecosystem.

Discussion

d. Implications

- The study highlights critical challenges that political communication and society face such as the profound interconnectedness of the hybrid and asymmetric media system, the deep political polarization, the attention-focused media logics, and the disinformation that can flourish in such an environment.
- Findings emphasize that “disinformation” cannot be easily separated from “legitimate” political expression in the polarized and fragmented media environment.
- Conservative media ecosystem is more vulnerable and susceptible to disinformation actors and their influence.

e. Limitations

- The study could not discern and separate potential internally-coordinated mechanisms such the IRA accounts following and/or retweeting each other to appear more authentic, engagement-worthy, and news-worthy
- The study examined only four Twitter accounts out of the thousands of IRA accounts.

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Response Letter (~600 words)

I think this study asks an innovative yet critical question: how disinformation actors gained social media followings, a crucial communication asset and power in the current media system. Beyond findings from previous studies that social media following plays a crucial role in broadening reach and amplifying influence, this study sheds light on the mechanisms underlying the construction of disinformation actors' social media following.

I liked that this study conceptualizes and explains that Twitter followers are not just the audience of the accounts they follow, but they function as a network that diffuses information to other like-minded social media users who don't follow the original source. I thought it was truly brilliant that they used variables related to "retweet" to capture this "ego-network" characteristic of followings. Retweet, which is the unique feature compared to legacy media like news media, allows information diffusion to nonfollowers by networks of followers on Twitter. Thus, I agree it was an appropriate variable to capture the network effect of Twitter following.

The most impressive strength is that this study investigated this mechanism by taking what they called a media system approach. In other words, the study sought to understand the mechanisms in reality, namely, an ideologically polarized, attention-driven, and asymmetric political communication system and society.

Another strength of this study is that they conducted the timeseries analysis using big data. I believe that its thorough timeseries analysis was an appropriate method addressing the hypotheses and the research question. More specifically, it is hard to claim causal relationships with this observational big data, since it is not from an experimental study. Yet, this study's timeseries analysis could demonstrate the temporal orders among variables.

However, data used in this study raises question that how much the findings have validity and implications. Examination of four most successful IRA accounts should not be the problem when the research goal is to understand how IRA disinformation social media achieve the success by gaining a large number of followers. The problem is that this study generalizes the findings from analyses using four accounts. For example, I don't think it is an overgeneralization media asymmetry between conservative and liberal media, just because conservative media embedded two IRA accounts used in the study, and liberal didn't use the two liberal IRA accounts used in the study.

Likewise, they used the sample from the Twitter followers of Donald Trump and a sample of Twitter users who retweeted the IRA accounts for their convenience. I believe this is a very problematic regarding representativeness. The study did not reveal the exact proportion of political stance of this data, simply stating the data was conservative-

leaning. Again, I don't think it is not proper to discuss asymmetry between conservative and liberal media based on findings from this biased data.

There could be other factors that contributed to the success of IRA social media accounts masquerading as conservatives compared to those masquerading as liberals. For example, claims and arguments of conservative faking IRA accounts could have been much more appealing to conservatives when liberal faking IRA accounts failed to meet the liberals' needs. Thus, we must be cautious about generalizing from biased or nonrepresentative samples, especially for specific topics or issues.

In sum, I think this study demonstrates the strength of quantitative case studies using big data and computational methods, revealing social media network effects in the context of the whole media ecosystem. These findings cannot be earned by conventional methods. However, although the sample should not always have to be a representative sample as long as it serves the research goal, we have to be cautious about the scope of the explanatory powers of the study.

References

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